

01 INTRODUCTION & OBJECTIVES

Operational setting & oversight bottleneck

US Navy submarine Fire Control Technicians monitor the combat system and cybersecurity display for a platform of nearly a thousand nodes. In this setting, AI-generated alerts and reasoning traces can arrive faster than any operator can review them, which drives alert fatigue and display disuse. [1]

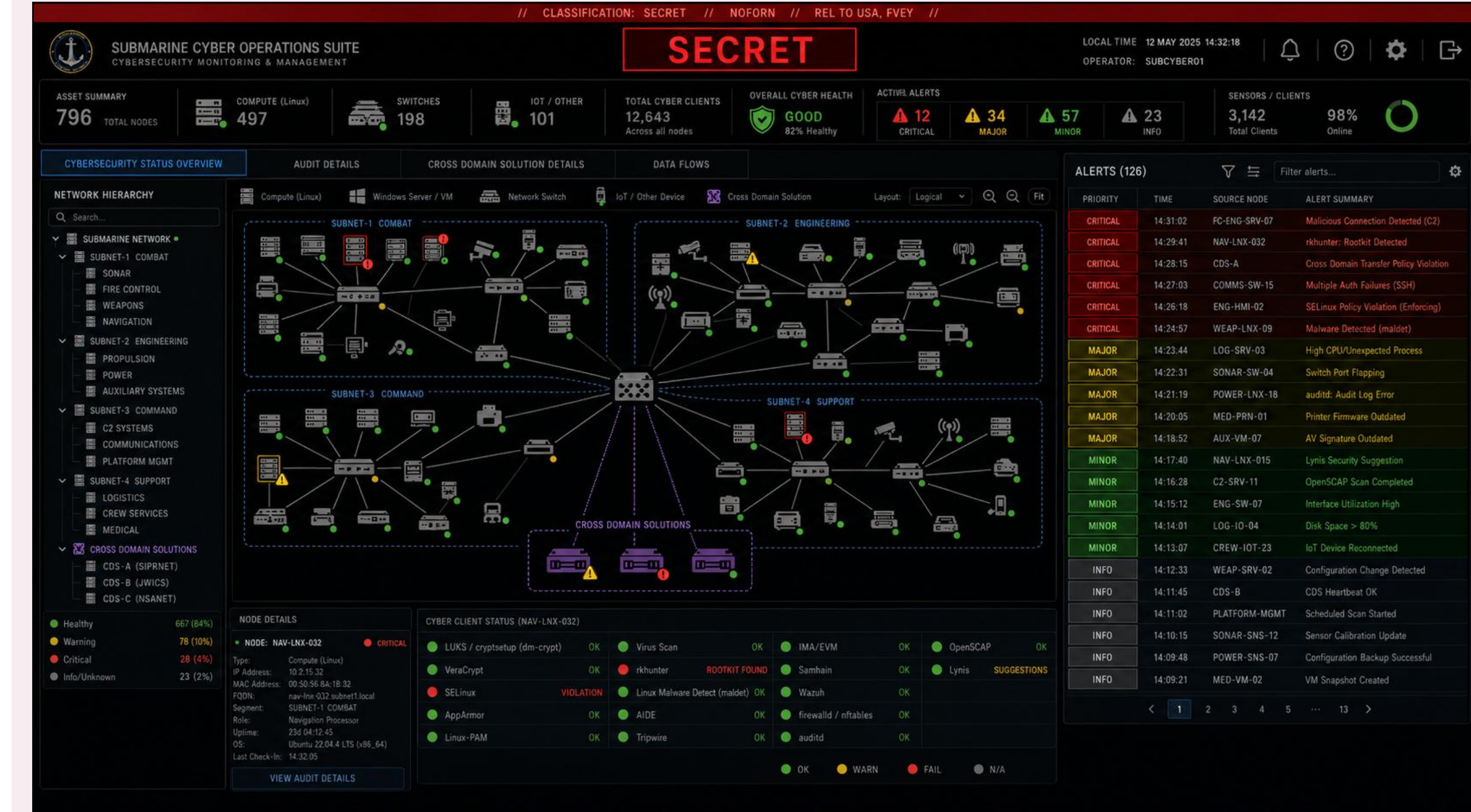


Figure 1. Cybersecurity Display shows network and system status, security posture, and active alerts. Figure is UNCLASSIFIED // DIST A. Prompt in Paper Appendix.

Aims & primary hypothesis

Estimate capacity from telemetry and allocate review effort jointly with risk and connectedness.

Hypothesis: environmental telemetry predicts measurable degradation in human review performance.

Method: platform as a sensor

CANDIDATE TELEMETRY RULES

Platform telemetry	Estimate
Self-protect states; hull openings	Higher load $L(t)$
Depth; contact density; maneuver tempo	Load $L(t)$
Training displays / simulated data	Low-load anchor
Active external interfaces	Exposure $E(t)$, not $L(t)$

02 PROPOSED FRAMEWORK



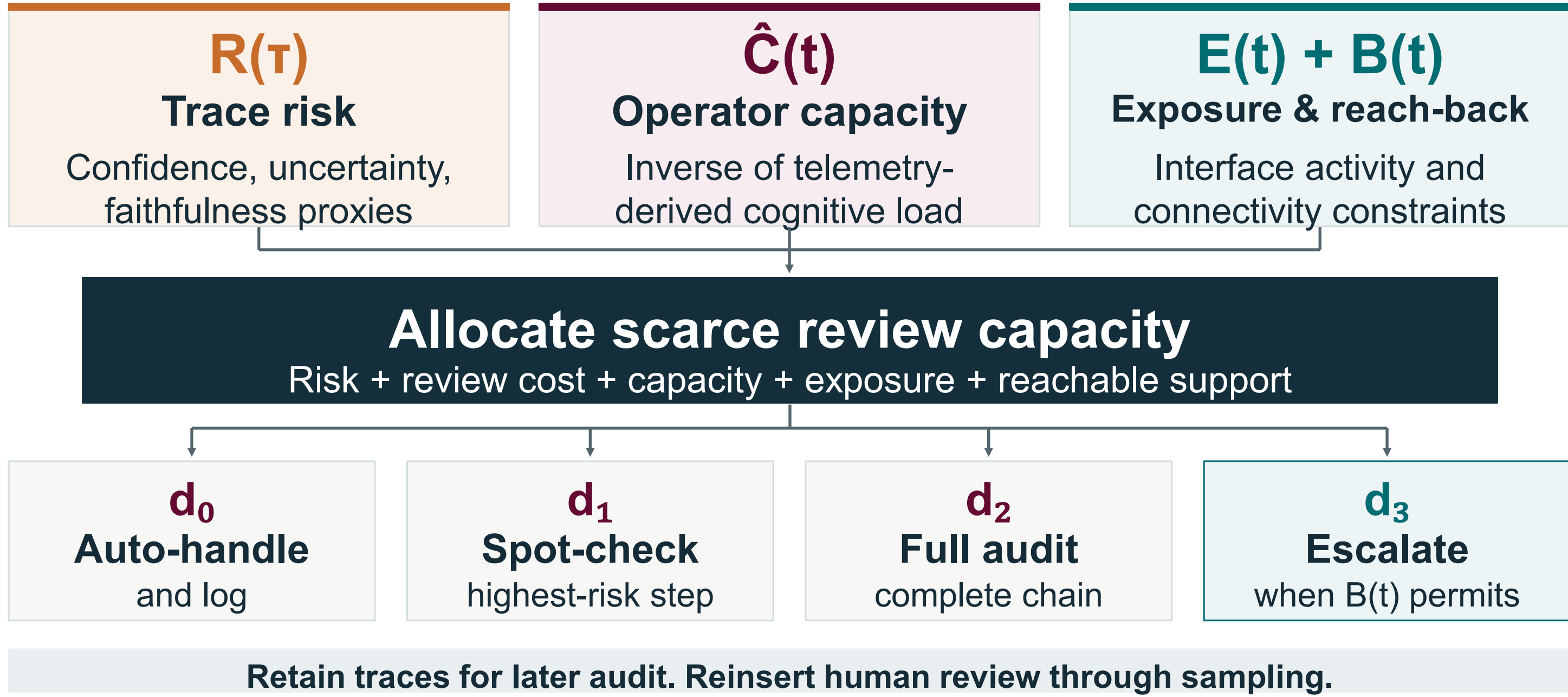
Operator Cognitive Load indicator (paper Fig. 2).

Load $L(t)$ maps inversely to capacity $\hat{C}(t)$.
No wearables or biometrics.

KEY INSIGHT:

Risk is not enough.
Match oversight to available human capacity.

Connectedness-aware oversight engine CONCEPTUAL ARCHITECTURE



Load and exposure are not interchangeable

Operational state	$L(t) / E(t)$	Routing implication
Routine communications window	Low / High	Lower escalation threshold.
Torpedo evasion; no external connectivity	High / Low	Defer low-value alerts.
Load and exposure both high	High / High	Use importance and $B(t)$ to suppress, interrupt, or escalate.

Candidate policy; telemetry rules and thresholds require calibration.

03 DISCUSSION & CONCLUSION

Designing for appropriate reliance

RELIANCE FAILURES

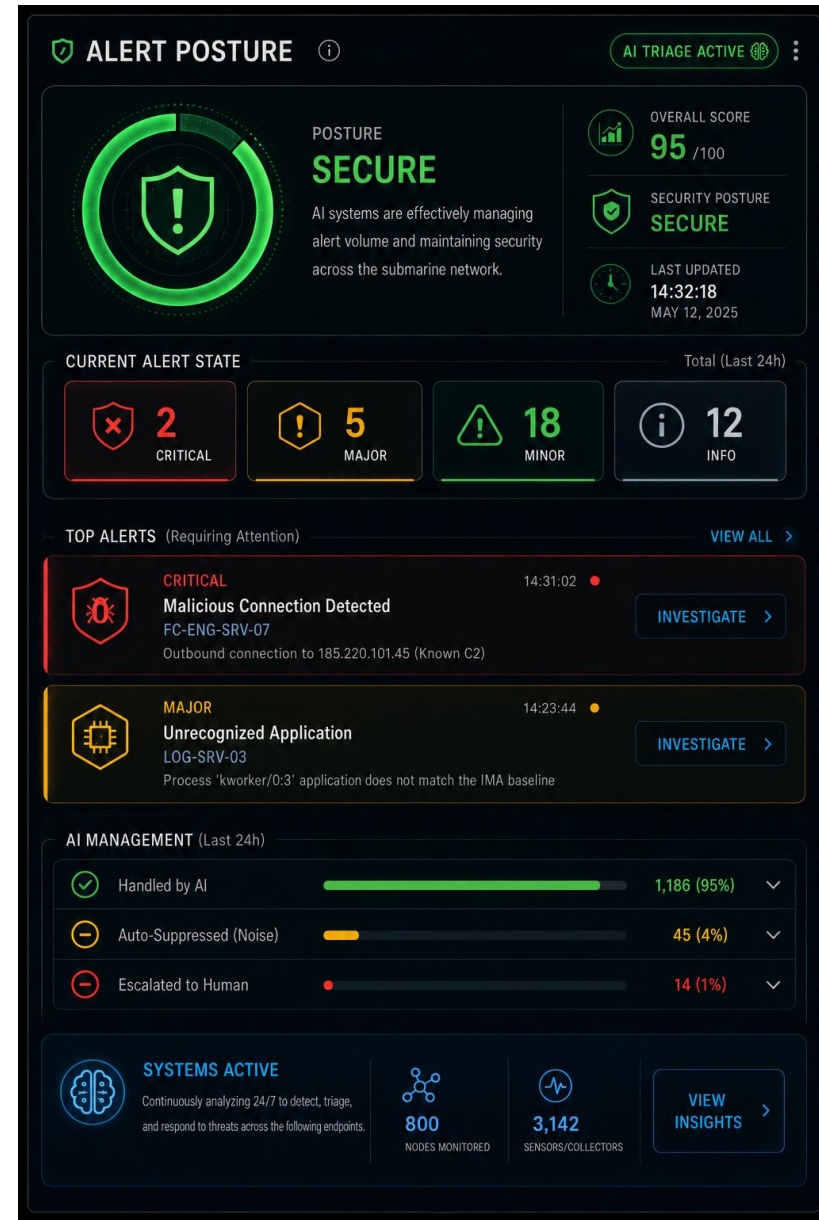
Disuse: repeated false alarms teach operators to ignore the display.

Over-reliance: adaptive automation can encourage complacent rubber-stamping. [2]

DESIGN REQUIREMENTS

Retain every auto-handled trace for audit.

Reinsert review through periodic forced-review sampling.



Illustrative interface (paper Fig. 4); displayed counts are not empirical results.

Conclusion & future evaluation

Oversight quality depends on available human capacity, not only on model risk.

Takeaways

Platform telemetry provides a non-invasive estimate of operator capacity.
Connectedness serves two roles: a load signal and a boundary on collective support.
The goal is appropriate reliance so the watchstander can still reason, believe, remember, judge, and act.

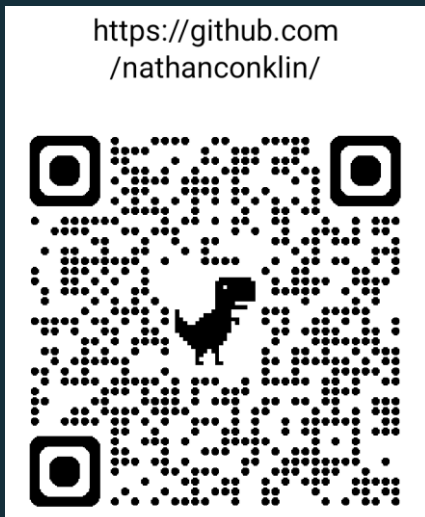
Evaluation focus

Test whether environmental context improves routing beyond model confidence.
Measure catch-rate-per-unit-capacity, miss rate, time-to-disposition, and calibrated reliance.
Check burden and equity so the mechanism does not relocate risk into an unauditable gate.

Design objective: preserve the conditions under which the human can perform meaningful oversight.

Acknowledgments

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DUAL ROLE OF CONNECTEDNESS

A load signal and a boundary on collective support.